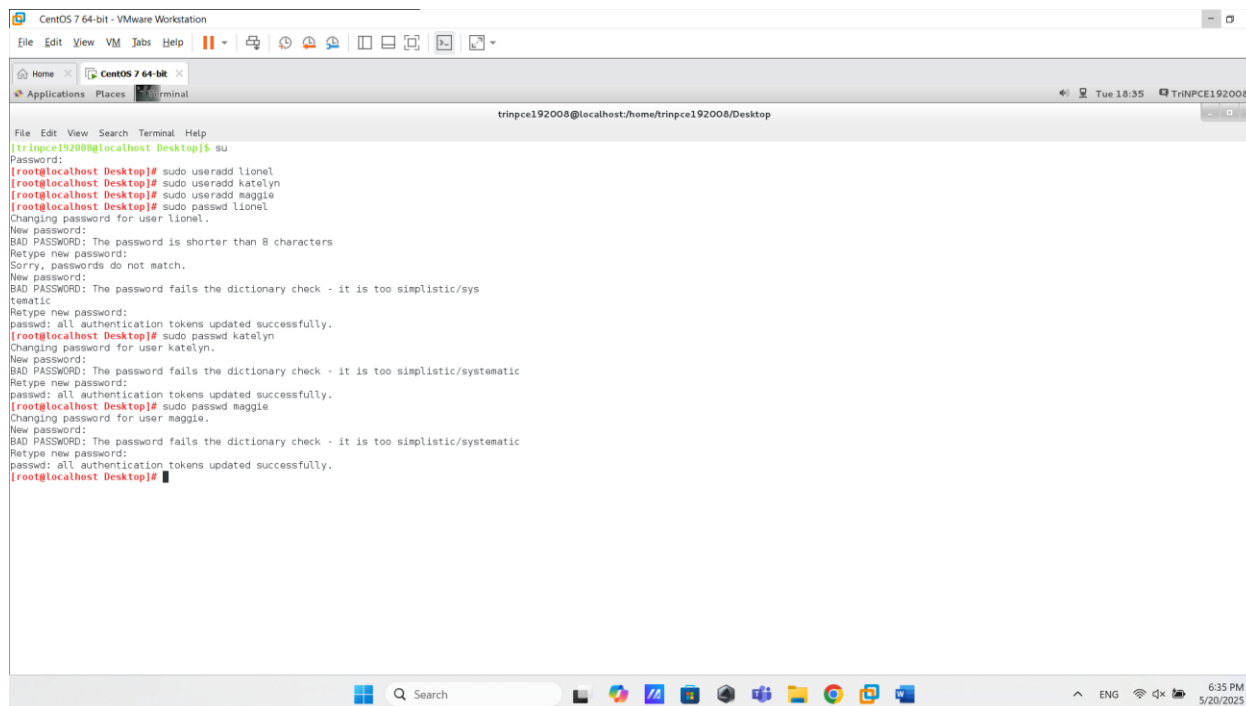


Hands-on lab for assigning limited sudo privileges

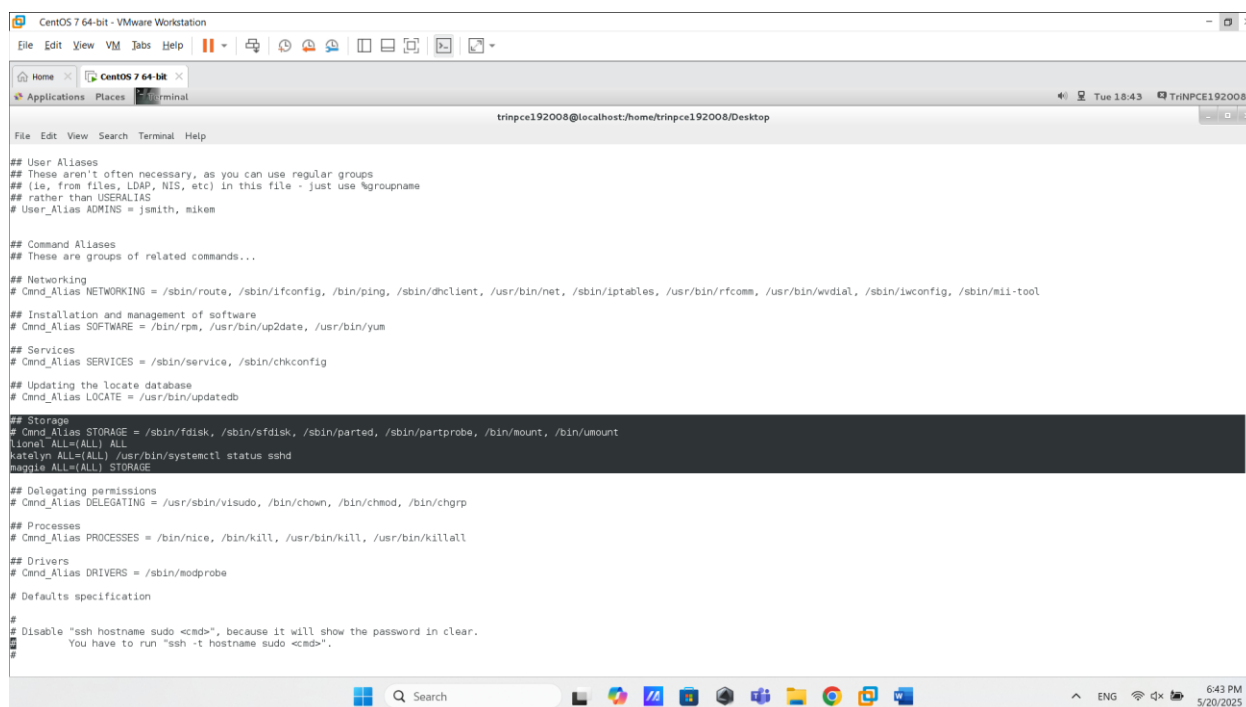
Tạo users:



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Applications Places Terminal
trinpc192008@localhost/home/trinpc192008/Desktop

trinpc192008@localhost Desktop]$ su
Password:
[root@localhost Desktop]# sudo useradd lionel
[root@localhost Desktop]# sudo useradd katelyn
[root@localhost Desktop]# sudo useradd maggie
[root@localhost Desktop]# sudo passwd lionel
Changing password for user lionel.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost Desktop]# sudo passwd katelyn
Changing password for user katelyn.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost Desktop]# sudo passwd maggie
Changing password for user maggie.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost Desktop]#
```

Cấp quyền bằng visudo:



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Applications Places Terminal
trinpc192008@localhost/home/trinpc192008/Desktop

File Edit View Search Terminal Help

## User Aliases
## These aren't often necessary, as you can use regular groups
## (ie, from files, LDAP, NIS, etc) in this file - just use %groupname
## rather than USERALIAS
# User_Alias ADMINS = jsmith, mikem

## Command Aliases
## These are groups of related commands...

## Networking
# Cmnd_Alias NETWORKING = /sbin/route, /sbin/ifconfig, /bin/ping, /sbin/dhclient, /usr/bin/net, /sbin/iptables, /usr/bin/rfcomm, /usr/bin/wvdial, /sbin/lwconfig, /sbin/mii-tool

## Installation and management of software
# Cmnd_Alias SOFTWARE = /bin/rpm, /usr/bin/up2date, /usr/bin/yum

## Services
# Cmnd_Alias SERVICES = /sbin/service, /sbin/chkconfig

## Updating the locate database
# Cmnd_Alias LOCATE = /usr/bin/updatedb

## Storage
# Cmnd_Alias STORAGE = /sbin/fdisk, /sbin/sfdisk, /sbin/parted, /sbin/partprobe, /bin/mount, /bin/umount
lionel ALL=(ALL) /usr/bin/systemctl status sshd
katelyn ALL=(ALL) /usr/bin/systemctl status sshd
maggie ALL=(ALL) STORAGE

## Delegating permissions
# Cmnd_Alias DELEGATING = /usr/sbin/visudo, /bin/chown, /bin/chmod, /bin/chgrp

## Processes
# Cmnd_Alias PROCESSES = /bin/nice, /bin/kill, /usr/bin/kill, /usr/bin/killall

## Drivers
# Cmnd_Alias DRIVERS = /sbin/modprobe

# Defaults specification
#
# Disable "ssh hostname sudo <cmd>", because it will show the password in clear.
# You have to run "ssh -t hostname sudo <cmd>".
#
```

```
File Edit View Search Terminal Help
[trinpce192008@localhost Desktop]# su
Password:
[root@localhost Desktop]# sudo useradd Lionel
[root@localhost Desktop]# sudo useradd katelyn
[root@localhost Desktop]# sudo useradd maggie
[root@localhost Desktop]# sudo passwd Lionel
Changing password for user Lionel.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/sys
tematic
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost Desktop]# sudo passwd katelyn
Changing password for user katelyn.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost Desktop]# sudo passwd maggie
Changing password for user maggie.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost Desktop]# clear
[trinpce192008@localhost Desktop]# visudo
visudo: Warning: Cnd Alias 'STORAGE' referenced but not defined
[root@localhost Desktop]# visudo
visudo: /etc/sudoers.tmp unchanged
visudo: Warning: Cnd Alias 'STORAGE' referenced but not defined
[root@localhost Desktop]# su - Lionel
[Lionel@localhost ~]$ sudo su -

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.
```

```
File Edit View Search Terminal Help
[root@localhost Desktop]# visudo
visudo: /etc/sudoers.tmp unchanged
visudo: Warning: Cnd Alias 'STORAGE' referenced but not defined
[root@localhost Desktop]# su - Lionel
[Lionel@localhost ~]$ sudo su -

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for Lionel:
Last login: Tue May 20 18:32:48 ICT 2025 on pts/0
[Lionel@localhost ~]$ exit
logout
[Lionel@localhost ~]$ sudo systemctl status sshd
sshd.service - OpenSSH server daemon
Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled)
Active: active (running) since Tue 2025-05-20 18:22:41 ICT; 8h ago
Process: 1517 ExecStartPre=/usr/sbin/sshd-keygen (code=exited, status=0/SUCCESS)
Main PID: 1523 (sshd)
CGroup: /system.slice/sshd.service
└─1523 /usr/sbin/sshd -D

May 20 18:22:41 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
May 20 18:22:41 localhost.localdomain sshd[1523]: Server listening on 0.0.0.0 port 22.
May 20 18:22:41 localhost.localdomain sshd[1523]: Server listening on :: port 22.
[Lionel@localhost ~]$ sudo fdisk -l

Disk /dev/sda: 21.5 GB, 21474836480 bytes, 41943040 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0x0008a90a

Device Boot      Start         End      Blocks   Id  System
/dev/sda1 *        2048       616447       307200    83  Linux
/dev/sda2          616448     41943039     20663296    8e  Linux LVM

Disk /dev/mapper/centos-swap: 2147 MB, 2147483648 bytes, 4194304 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
```

Ảnh trên: chạy 1 số lệnh để xác thực quyền của user lionel

```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Applications Places Terminal
trinpce192008@localhost/home/trinpce192008/Desktop

File Edit View Search Terminal Help
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 65536 bytes

Disk /dev/mapper/centos-pool08: 15.8 GB, 15837691904 bytes, 38932992 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 65536 bytes

[trinpce192008@localhost ~]$ exit
logout
[root@localhost Desktop]# su - katelyn
[katelyn@localhost ~]$ sudo su -

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for katelyn:
Sorry, user katelyn is not allowed to execute '/bin/su -' as root on localhost.localdomain.
[katelyn@localhost ~]$ sudo systemctl status sshd
[sudo] password for katelyn:
Sorry, try again.
[sudo] password for katelyn:
sshd.service - OpenSSH server daemon
Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled)
Active: active (running) since Tue 2025-05-20 18:22:41 ICT; 8h ago
Process: 1517 ExecStartPre=/usr/sbin/sshd-keygen (code=exited, status=0/SUCCESS)
Main PID: 1523 (sshd)
CGroup: /system.slice/sshd.service
└─1523 /usr/sbin/sshd -D

May 20 18:22:41 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
May 20 18:22:41 localhost.localdomain sshd[1523]: Server listening on 0.0.0.0 port 22.
May 20 18:22:41 localhost.localdomain sshd[1523]: Server listening on :: port 22.
[katelyn@localhost ~]$ sudo systemctl restart sshd
Sorry, user katelyn is not allowed to execute '/bin/systemctl restart sshd' as root on localhost.localdomain.
[katelyn@localhost ~]$ sudo fdisk -l
Sorry, user katelyn is not allowed to execute '/sbin/fdisk -l' as root on localhost.localdomain.
[katelyn@localhost ~]$ exit
logout
[root@localhost Desktop]#
```

```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Applications Places Terminal
trinpce192008@localhost/home/trinpce192008/Desktop

File Edit View Search Terminal Help
[katelyn@localhost ~]$ sudo systemctl status sshd
[sudo] password for katelyn:
Sorry, try again.
[sudo] password for katelyn:
sshd.service - OpenSSH server daemon
Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled)
Active: active (running) since Tue 2025-05-20 18:22:41 ICT; 8h ago
Process: 1517 ExecStartPre=/usr/sbin/sshd-keygen (code=exited, status=0/SUCCESS)
Main PID: 1523 (sshd)
CGroup: /system.slice/sshd.service
└─1523 /usr/sbin/sshd -D

May 20 18:22:41 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
May 20 18:22:41 localhost.localdomain sshd[1523]: Server listening on 0.0.0.0 port 22.
May 20 18:22:41 localhost.localdomain sshd[1523]: Server listening on :: port 22.
[katelyn@localhost ~]$ sudo systemctl restart sshd
Sorry, user katelyn is not allowed to execute '/bin/systemctl restart sshd' as root on localhost.localdomain.
[katelyn@localhost ~]$ sudo fdisk -l
Sorry, user katelyn is not allowed to execute '/sbin/fdisk -l' as root on localhost.localdomain.
[katelyn@localhost ~]$ exit
logout
[root@localhost Desktop]# su - maggie
[maggie@localhost ~]$ sudo su -

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for maggie:
Sorry, user maggie is not allowed to execute '/bin/su -' as root on localhost.localdomain.
[maggie@localhost ~]$ sudo systemctl status sshd
[sudo] password for maggie:
Sorry, user maggie is not allowed to execute '/bin/systemctl status sshd' as root on localhost.localdomain.
[maggie@localhost ~]$ sudo systemctl restart sshd
[sudo] password for maggie:
Sorry, user maggie is not allowed to execute '/bin/systemctl restart sshd' as root on localhost.localdomain.
[maggie@localhost ~]$ sudo fdisk -l
[sudo] password for maggie:
Sorry, user maggie is not allowed to execute '/sbin/fdisk -l' as root on localhost.localdomain.
[maggie@localhost ~]$ exit
logout
[root@localhost Desktop]#
```

Hands-on lab for disabling the sudo timer

```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Applications Places Terminal
trinpce192008@localhost:~$ sudo fdisk -l
[sudo] password for trinpce192008:
Disk /dev/sda: 21.5 GB, 21474836480 bytes, 41943840 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0x0008a90a

   Device Boot      Start         End      Blocks   Id  System
/dev/sda1 *        2048        616447     307200    83  Linux
/dev/sda2          616448    41943039    20663296    8e  Linux LVM

Disk /dev/mapper/centos-swap: 2147 MB, 2147483648 bytes, 4194304 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/centos-pool08_tmeta: 16 MB, 16777216 bytes, 32768 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/centos-pool08_tdata: 15.8 GB, 15837691904 bytes, 30932992 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/centos-pool08-tpool: 15.8 GB, 15837691904 bytes, 30932992 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 65536 bytes

Disk /dev/mapper/centos-root: 5284 MB, 5284823040 bytes, 10321920 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 65536 bytes
```

```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Applications Places Terminal
trinpce192008@localhost:~$ clear
trinpce192008@localhost ~$ sudo systemctl status sshd
sshd.service - OpenSSH server daemon
Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled)
Active: active (running) since Tue 2025-05-20 10:22:41 ICT; 8h ago
Process: 1517 ExecStartPre=/usr/sbin/ssh-keygen (code=exited, status=0/SUCCESS)
Main PID: 1523 (sshd)
CGroup: /system.slice/ssh.service
└─1523 /usr/sbin/sshd -D

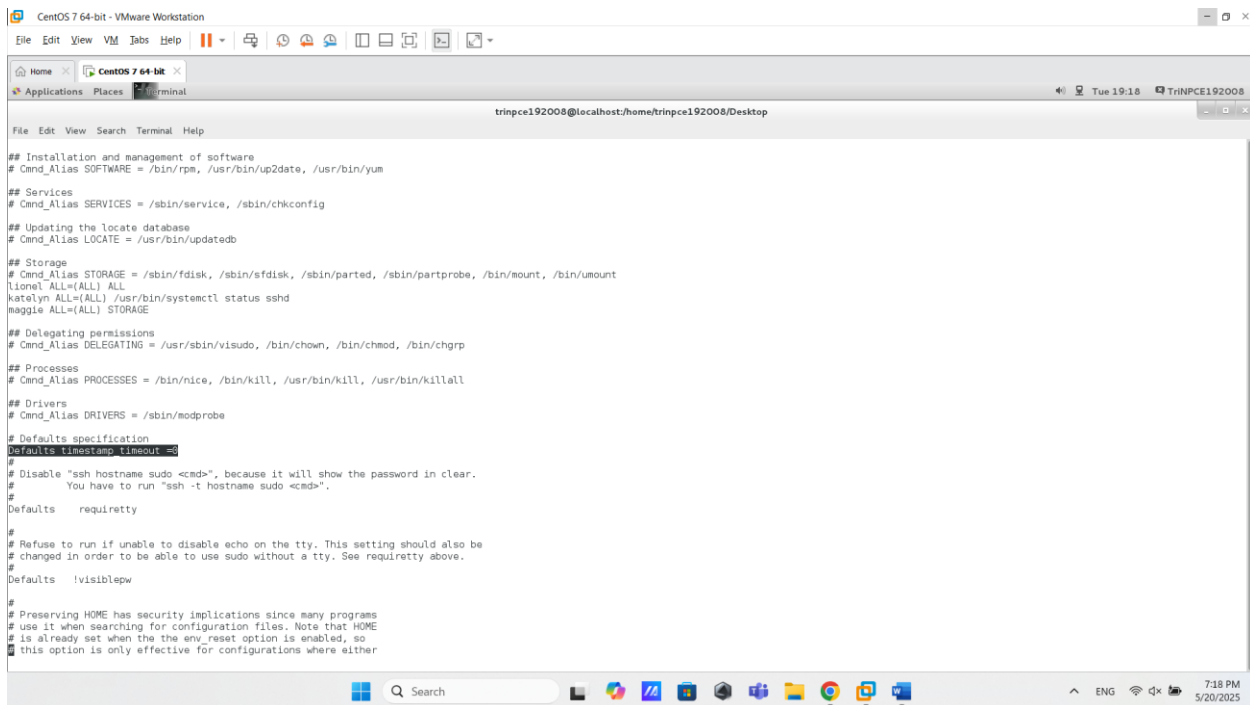
May 20 10:22:41 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
May 20 10:22:41 localhost.localdomain sshd[1523]: Server listening on 0.0.0.0 port 22.
May 20 10:22:41 localhost.localdomain sshd[1523]: Server listening on :: port 22.
trinpce192008@localhost ~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target prot opt source destination
ACCEPT all -- anywhere anywhere ctstate RELATED,ESTABLISHED
ACCEPT all -- anywhere anywhere
INPUT_direct all -- anywhere anywhere
INPUT_ZONES_SOURCE all -- anywhere anywhere
INPUT_ZONES all -- anywhere anywhere
ACCEPT icmp -- anywhere anywhere
REJECT all -- anywhere anywhere reject-with icmp-host-prohibited

Chain FORWARD (policy ACCEPT)
target prot opt source destination
ACCEPT all -- anywhere anywhere ctstate RELATED,ESTABLISHED
ACCEPT all -- anywhere anywhere
FORWARD_direct all -- anywhere anywhere
FORWARD_IN_ZONES_SOURCE all -- anywhere anywhere
FORWARD_IN_ZONES all -- anywhere anywhere
FORWARD_OUT_ZONES_SOURCE all -- anywhere anywhere
FORWARD_OUT_ZONES all -- anywhere anywhere
ACCEPT icmp -- anywhere anywhere
REJECT all -- anywhere anywhere reject-with icmp-host-prohibited

Chain OUTPUT (policy ACCEPT)
target prot opt source destination
OUTPUT_direct all -- anywhere anywhere

Chain FORWARD_IN_ZONES (1 references)
target prot opt source destination
FWDI_public all -- anywhere anywhere [goto]
FWDI_public all -- anywhere anywhere [goto]
```

Ảnh trên: không cần nhập mật khẩu mỗi lần chạy lệnh



```
## Installation and management of software
# Cnmd_Alias SOFTWARE = /bin/rpm, /usr/bin/up2date, /usr/bin/yum

## Services
# Cnmd_Alias SERVICES = /sbin/service, /sbin/chkconfig

## Updating the locate database
# Cnmd_Alias LOCATE = /usr/bin/updatedb

## Storage
# Cnmd_Alias STORAGE = /sbin/fdisk, /sbin/sfdisk, /sbin/parted, /sbin/partprobe, /bin/mount, /bin/umount
lione1 ALL=(ALL) ALL
katelyn ALL=(ALL) /usr/bin/systemctl status sshd
maggie ALL=(ALL) STORAGE

## Delegating permissions
# Cnmd_Alias DELEGATING = /usr/sbin/visudo, /bin/chown, /bin/chmod, /bin/chgrp

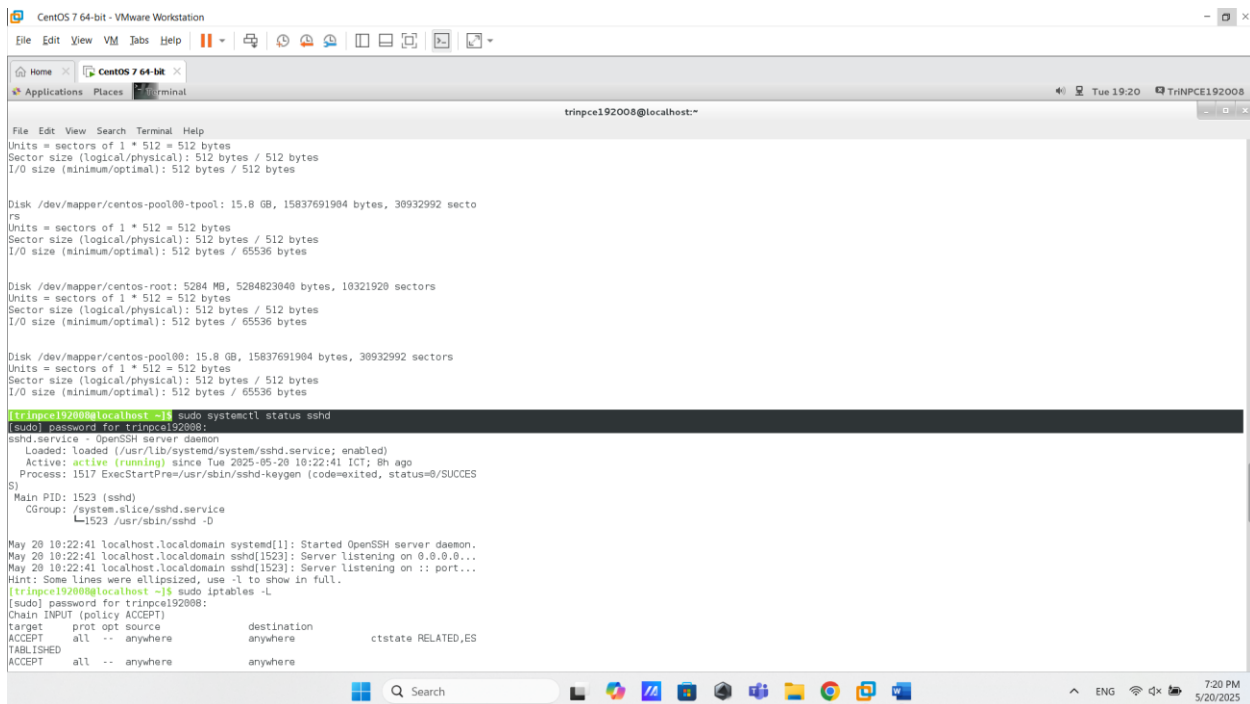
## Processes
# Cnmd_Alias PROCESSES = /bin/nice, /bin/kill, /usr/bin/kill, /usr/bin/killall

## Drivers
# Cnmd_Alias DRIVERS = /sbin/modprobe

# Defaults specification
Defaults timestamp_timeout=0
#
# Disable "ssh hostname sudo <cmd>", because it will show the password in clear.
# You have to run "ssh -t hostname sudo <cmd>".
#
Defaults requiretty

#
# Refuse to run if unable to disable echo on the tty. This setting should also be
# changed in order to be able to use sudo without a tty. See requiretty above.
#
Defaults !visiblepw

#
# Preserving HOME has security implications since many programs
# use it when searching for configuration files. Note that HOME
# is already set when the the env_reset option is enabled, so
# this option is only effective for configurations where either
```



```
File Edit View Search Terminal Help
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/centos-pool00-tpool: 15.8 GB, 15837691904 bytes, 30932992 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 65536 bytes

Disk /dev/mapper/centos-root: 5284 MB, 5284823040 bytes, 10321920 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 65536 bytes

Disk /dev/mapper/centos-pool00: 15.8 GB, 15837691904 bytes, 30932992 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 65536 bytes

[trinpce192008@localhost ~]$ sudo systemctl status sshd
[sudo] password for trinpce192008:
sshd.service - OpenSSH server daemon
Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled)
Active: active (running) since Tue 2025-05-20 10:22:41 ICT; 8m ago
Process: 1517 ExecStartPre=/usr/sbin/sshd-keygen (code=exited, status=0/SUCCESS)
Main PID: 1523 (sshd)
CGroup: /system.slice/sshd.service
└─1523 /usr/sbin/sshd -D

May 20 10:22:41 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
May 20 10:22:41 localhost.localdomain sshd[1523]: Server listening on 0.0.0.0...
May 20 10:22:41 localhost.localdomain sshd[1523]: Server listening on :: port...
Hint: Some lines were ellipsized, use -l to show in full.
[trinpce192008@localhost ~]$ sudo iptables -L
[sudo] password for trinpce192008:
Chain INPUT (policy ACCEPT)
target prot opt source destination ctstate RELATED,ES
ACCEPT all -- anywhere anywhere
ESTABLISHED
ACCEPT all -- anywhere anywhere
```

2 Ảnh trên: sau khi thêm Defaults timestamp_timeout=0 phải nhập mật khẩu mỗi lần chạy lệnh

```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 7 64-bit
Applications Places Terminal
trinpce192008@localhost/home/trinpce192008/Desktop

File Edit View Search Terminal Help
# Cmd_Alias DELEGATING = /usr/sbin/visudo, /bin/chown, /bin/chmod, /bin/chgrp

## Processes
# Cmd_Alias PROCESSES = /bin/nice, /bin/kill, /usr/bin/kill, /usr/bin/killall

## Drivers
# Cmd_Alias DRIVERS = /sbin/modprobe

# Defaults specification
Defaults:lionel timestamp timeout = 0

#
# Disable "ssh hostname sudo <cmd>", because it will show the password in clear.
# You have to run "ssh -t hostname sudo <cmd>".
# Defaults    requiretty

#
# Refuse to run if unable to disable echo on the tty. This setting should also be
# changed in order to be able to use sudo without a tty. See requiretty above.
# Defaults    !visiblepw

#
# Preserving HOME has security implications since many programs
# use it when searching for configuration files. Note that HOME
# is already set when the the env_reset option is enabled, so
# this option is only effective for configurations where either
# env_reset is disabled or HOME is present in the env_keep list.
# Defaults    always_set_home

Defaults    env_reset
Defaults    env_keep += "COLORS DISPLAY HOSTNAME HISTSIZE INPUTRC KDEDIR LS_COLORS"
Defaults    env_keep += "MAIL PS1 PS2 QTDIR USERNAME LANG LC_ADDRESS LC_CTYPE"
Defaults    env_keep += "LC_COLLATE LC_IDENTIFICATION LC_MEASUREMENT LC_MESSAGES"
Defaults    env_keep += "LC_MONETARY LC_NAME LC_NUMERIC LC_PAPER LC_TELEPHONE"
Defaults    env_keep += "LC_TIME LC_ALL LANGUAGE LINGUAS _XKB_CHARSET AUTHORITY"

#
# Adding HOME to env_keep may enable a user to run unrestricted
# commands via sudo.
#
# Defaults    env_keep += "HOME"
-- INSERT --
```

```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 7 64-bit
Applications Places Terminal
trinpce192008@localhost~

File Edit View Search Terminal Help
[trinpce192008@localhost ~]$ sudo fdisk -l
[sudo] password for trinpce192008:

Disk /dev/sda: 21.5 GB, 21474836480 bytes, 41943040 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0x0008a90a

   Device Boot      Start         End      Blocks   Id  System
/dev/sda1 *         2048        616447       307200    83  Linux
/dev/sda2            616448    41943039    20663296    8e  Linux LVM

Disk /dev/mapper/centos-swap: 2147 MB, 2147483648 bytes, 4194304 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/centos-pool00_tmeta: 16 MB, 16777216 bytes, 32768 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/centos-pool00_tdata: 15.8 GB, 15837691904 bytes, 30932992 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/centos-pool00-tpool: 15.8 GB, 15837691904 bytes, 30932992 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

trinpce192008@localhost~

File Edit View Search Terminal Help
Password:
Last login: Tue May 20 18:44:52 ICT 2025 on pts/0
lionel@localhost:~$ sudo fdisk -l
[sudo] password for Lionel:

Disk /dev/sda: 21.5 GB, 21474836480 bytes, 41943040 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0x0008a90a

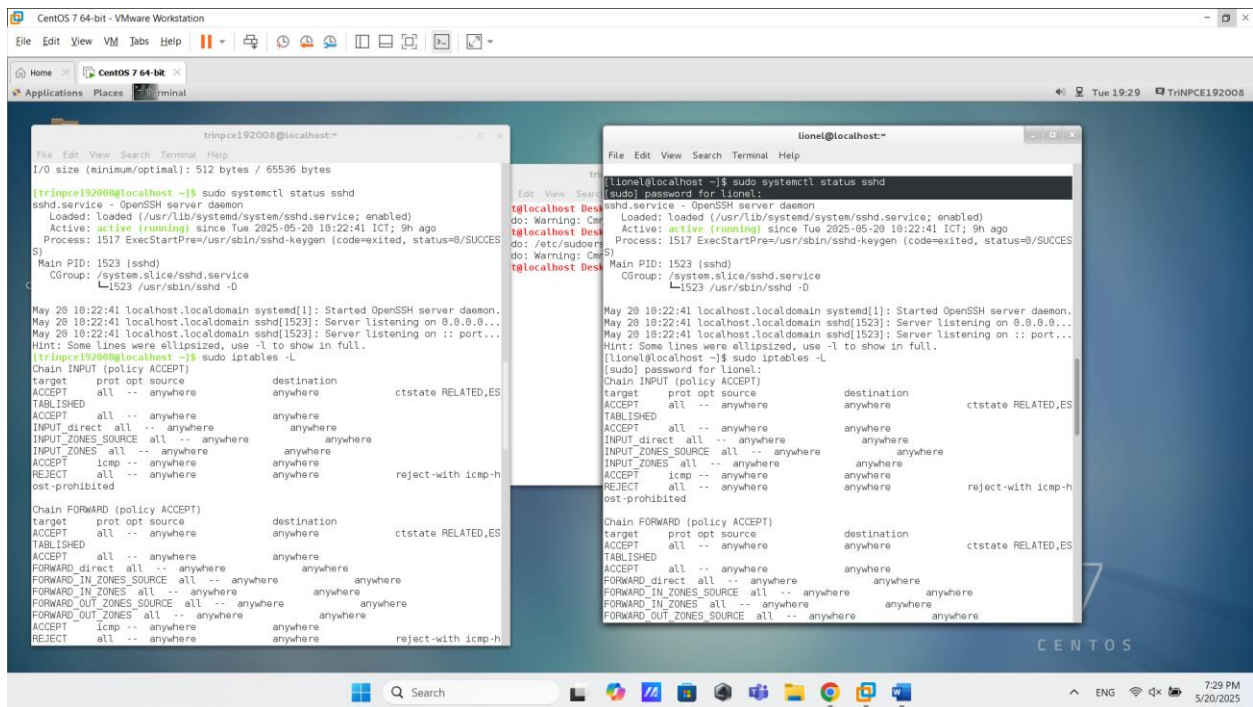
   Device Boot      Start         End      Blocks   Id  System
/dev/sda1 *         2048        616447       307200    83  Linux
/dev/sda2            616448    41943039    20663296    8e  Linux LVM

Disk /dev/mapper/centos-swap: 2147 MB, 2147483648 bytes, 4194304 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/mapper/centos-pool00_tmeta: 16 MB, 16777216 bytes, 32768 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

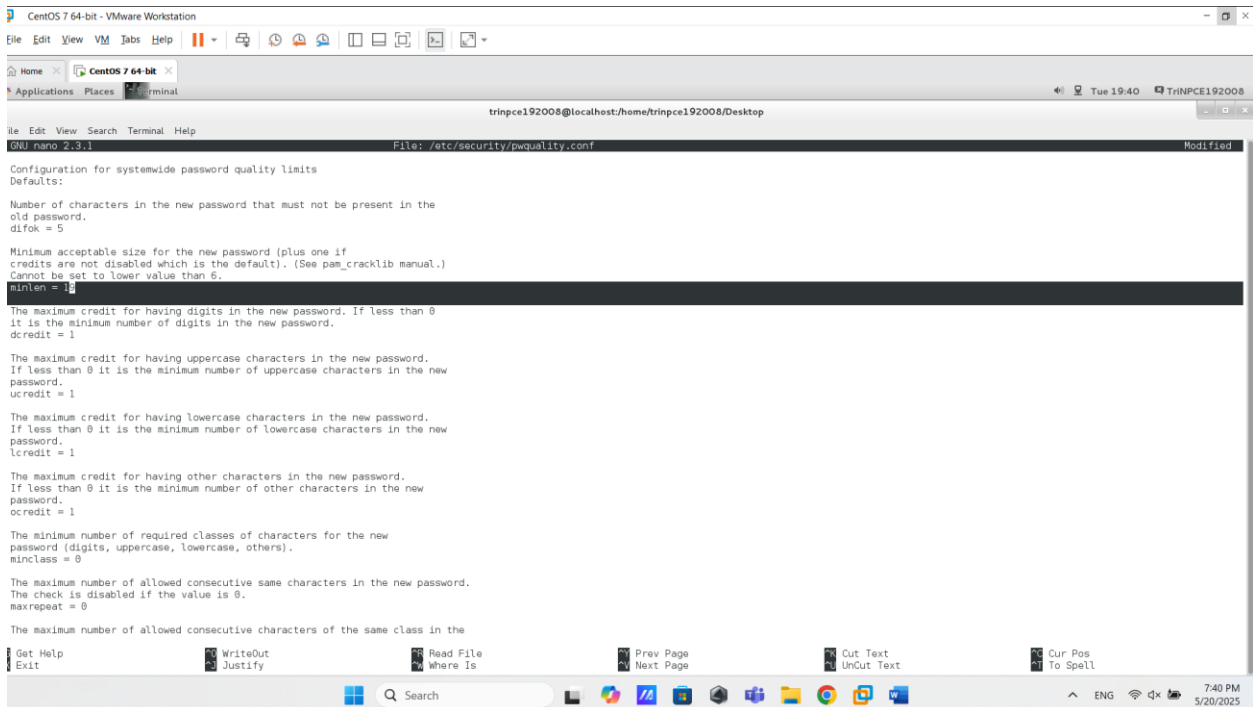
Disk /dev/mapper/centos-pool00_tdata: 15.8 GB, 15837691904 bytes, 30932992 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

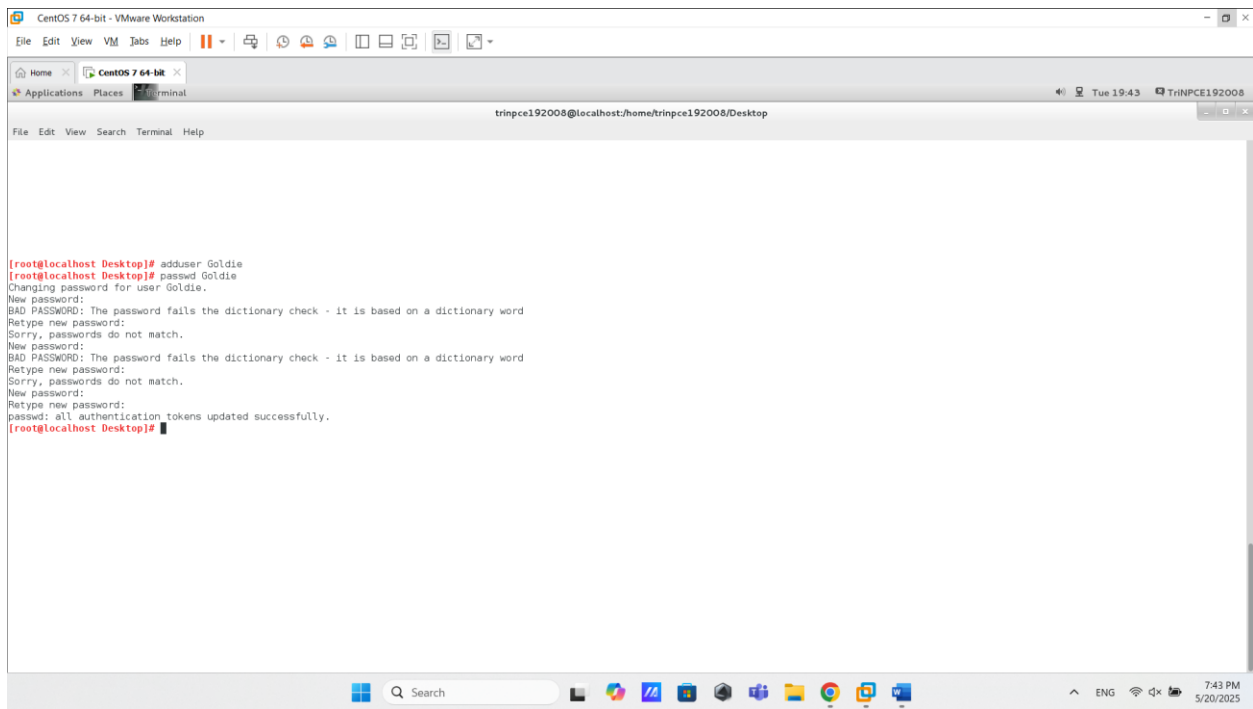
Disk /dev/mapper/centos-pool00-tpool: 15.8 GB, 15837691904 bytes, 30932992 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```



3 ảnh trên: sau khi thêm dòng Defaults:lionel timestamp_timeout = 0 thì chỉ có user lionel phải nhập mật khẩu mỗi lần chạy lệnh

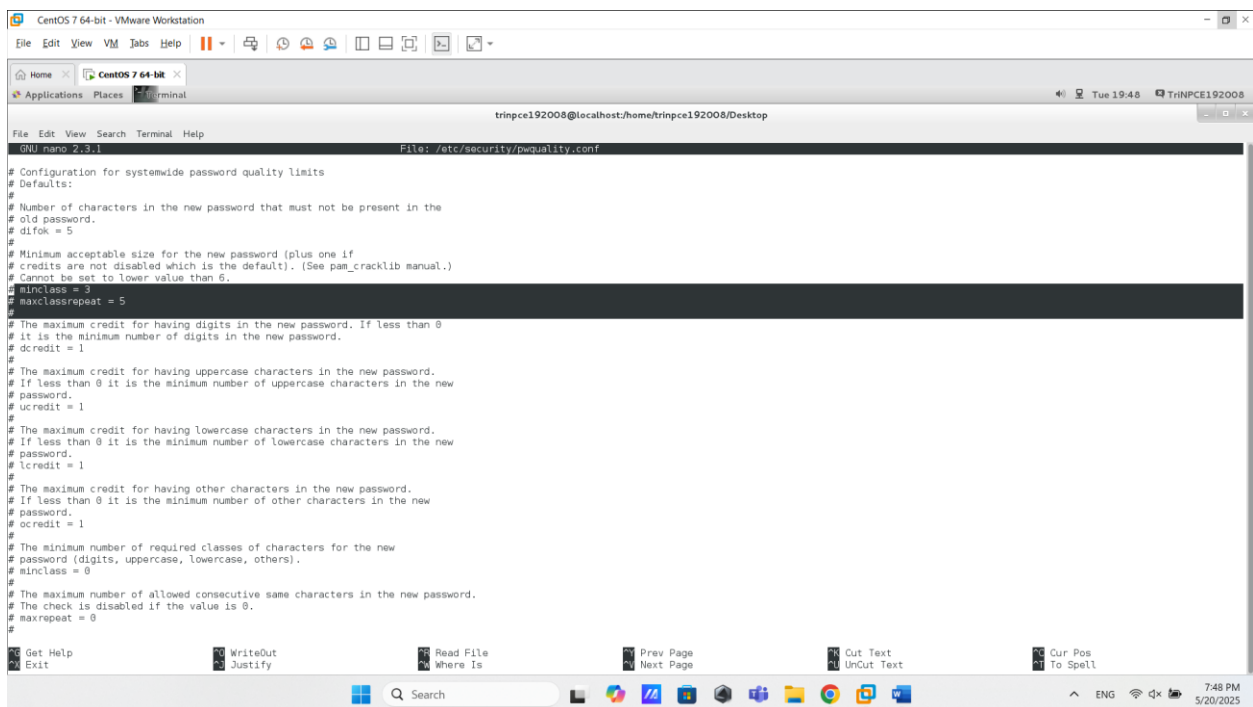
Hands-on lab for setting password complexity criteria



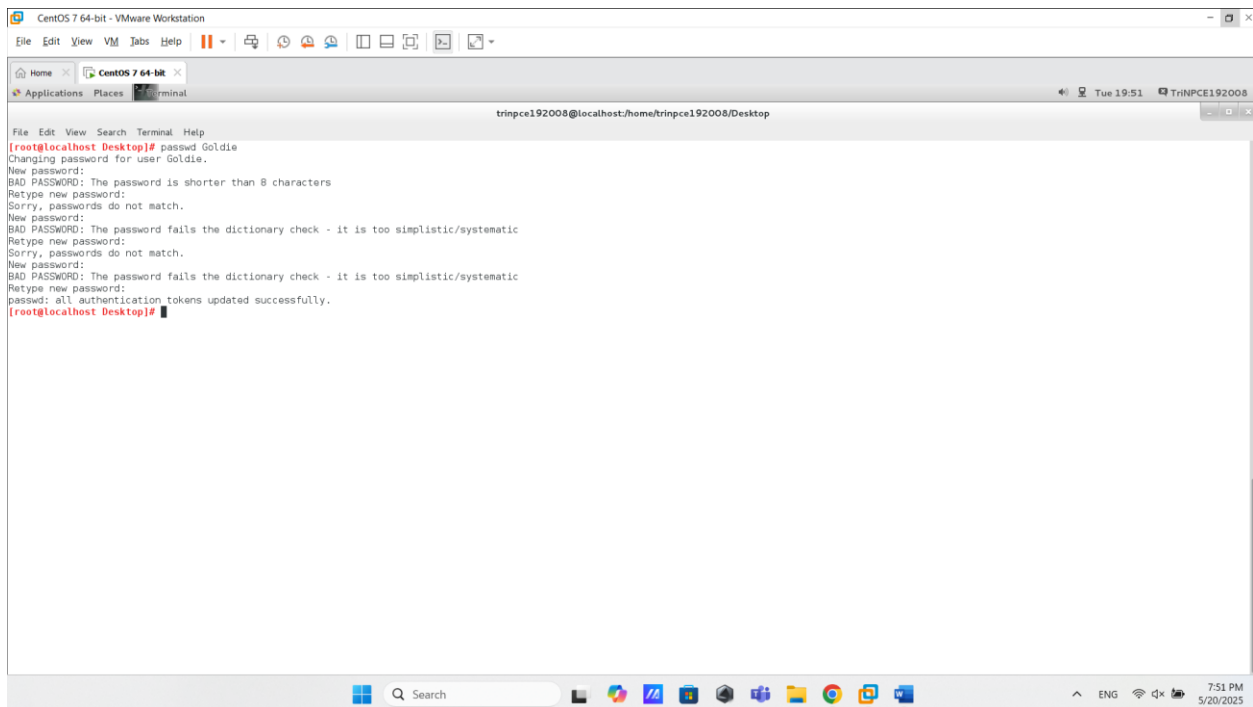


```
[root@localhost Desktop]# adduser Goldie
[root@localhost Desktop]# passwd Goldie
Changing password for user Goldie.
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
Sorry, passwords do not match.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost Desktop]#
```

Ảnh trên: sửa file thành minlen = 19 và yêu cầu đặt password cho user sẽ thay đổi



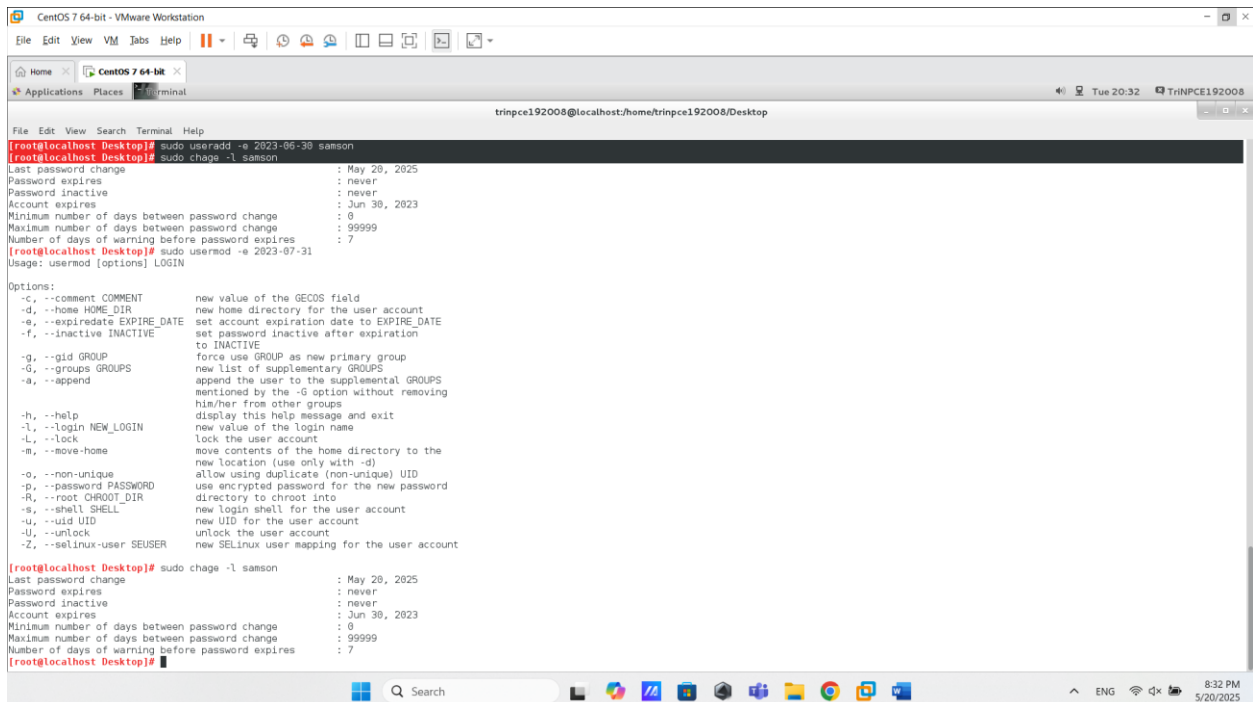
```
GNU nano 2.3.1 File: /etc/security/pwquality.conf
# Configuration for systemwide password quality limits
# Defaults:
#
# Number of characters in the new password that must not be present in the
# old password.
# difok = 5
#
# Minimum acceptable size for the new password (plus one if
# credits are not disabled which is the default). (See pam_cracklib manual.)
# Cannot be set to lower value than 6.
# minclass = 3
# maxclassrepeat = 5
#
# The maximum credit for having digits in the new password. If less than 0
# it is the minimum number of digits in the new password.
# dcredit = 1
#
# The maximum credit for having uppercase characters in the new password.
# If less than 0 it is the minimum number of uppercase characters in the new
# password.
# ucredit = 1
#
# The maximum credit for having lowercase characters in the new password.
# If less than 0 it is the minimum number of lowercase characters in the new
# password.
# lcredit = 1
#
# The maximum credit for having other characters in the new password.
# If less than 0 it is the minimum number of other characters in the new
# password.
# ocredit = 1
#
# The minimum number of required classes of characters for the new
# password (digits, uppercase, lowercase, others).
# minclass = 0
#
# The maximum number of allowed consecutive same characters in the new password.
# The check is disabled if the value is 0.
# maxrepeat = 0
#
Get Help      WriteOut    Read File    Prev Page    Cut Text     Cur Pos
Exit          Justify     Where Is     Next Page    UnCut Text   To Spell
```



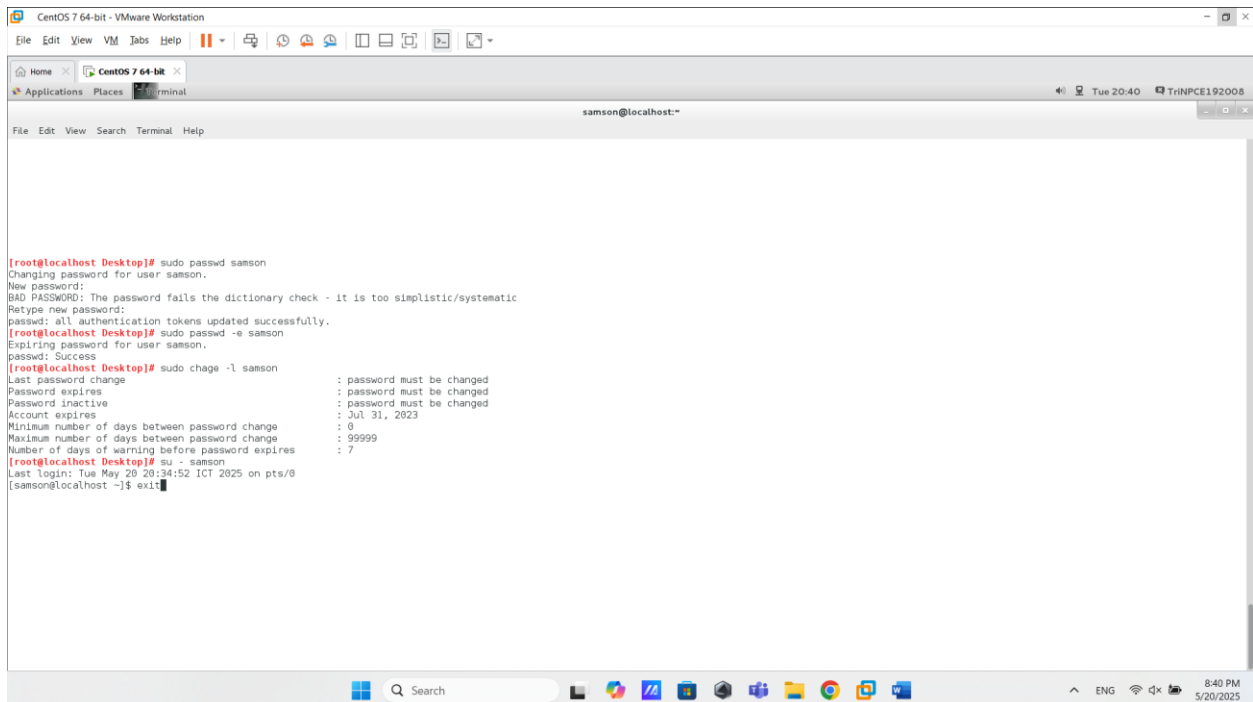
2 ảnh trên: thay đổi yêu cầu độ phức tạp khi đặt mật khẩu cho user

.....

Hands-on lab for setting account and password expiry data

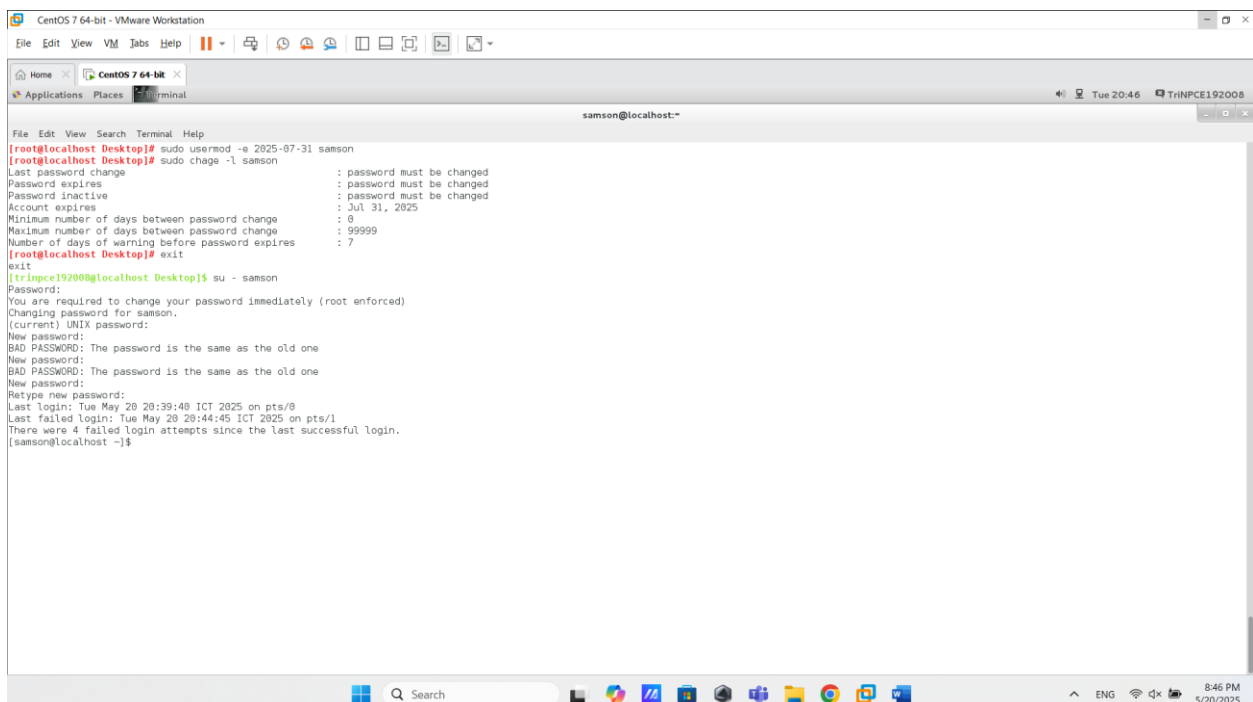


ảnh trên: tạo user samson và đặt ngày hết hạn cho user là 30-07-2023



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Applications Places Terminal
samson@localhost:~$

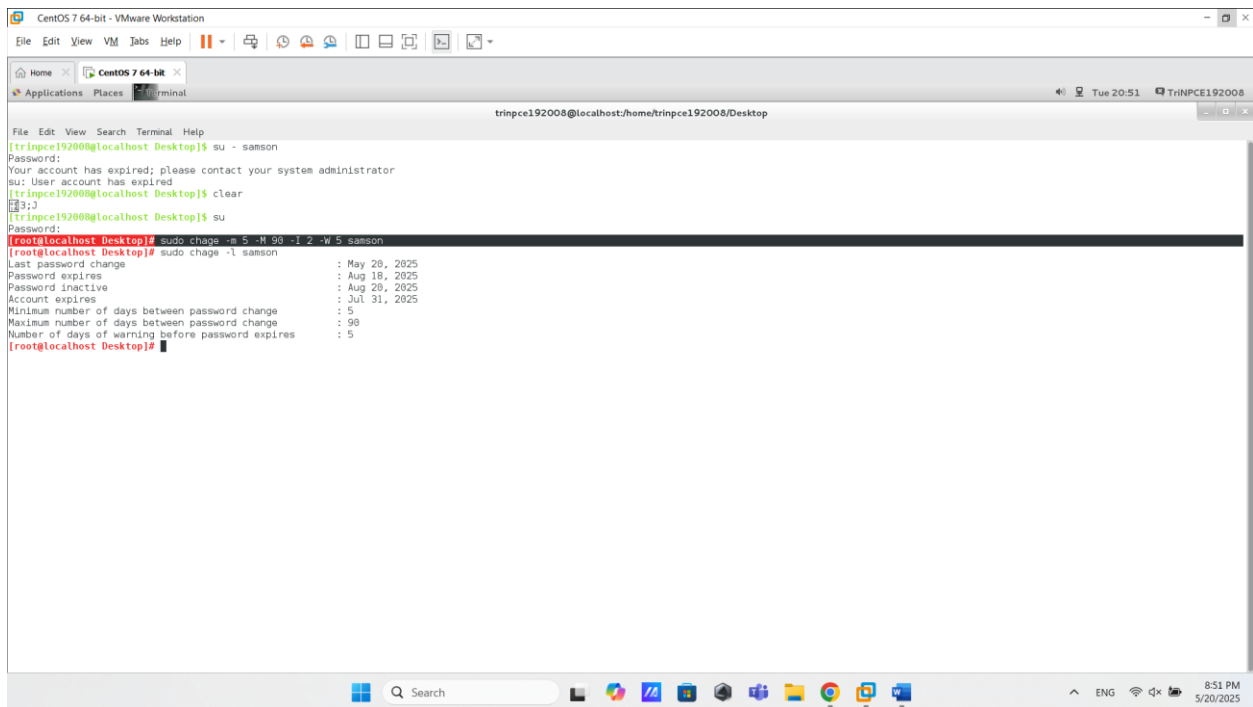
[root@localhost Desktop]# sudo passwd samson
Changing password for user samson.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost Desktop]# sudo passwd -e samson
Expiring password for user samson.
passwd: Success
[root@localhost Desktop]# sudo chage -l samson
Last password change           : password must be changed
Password expires               : password must be changed
Password inactive              : password must be changed
Account expires                : Jul 31, 2023
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
[root@localhost Desktop]# su - samson
Last login: Tue May 20 20:34:52 ICT 2025 on pts/0
[samson@localhost ~]$ exit
```



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Applications Places Terminal
samson@localhost:~$

[root@localhost Desktop]# sudo usermod -e 2025-07-31 samson
[root@localhost Desktop]# sudo chage -l samson
Last password change           : password must be changed
Password expires               : password must be changed
Password inactive              : password must be changed
Account expires                : Jul 31, 2025
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
[root@localhost Desktop]# exit
exit
[trinpce192008@localhost Desktop]$ su - samson
Password:
You are required to change your password immediately (root enforced)
Changing password for samson.
(current) UNIX password:
New password:
BAD PASSWORD: The password is the same as the old one
New password:
BAD PASSWORD: The password is the same as the old one
New password:
Retype new password:
Last login: Tue May 20 20:39:40 ICT 2025 on pts/0
Last failed login: Tue May 20 20:44:45 ICT 2025 on pts/1
There were 4 failed login attempts since the last successful login.
[samson@localhost ~]$
```

ảnh trên: thay đổi ngày hết hạn cho user samson là 31-07-2025, thay đổi mật khẩu của samson và bắt buộc user samson phải đổi mật khẩu vào lần đăng nhập đầu tiên



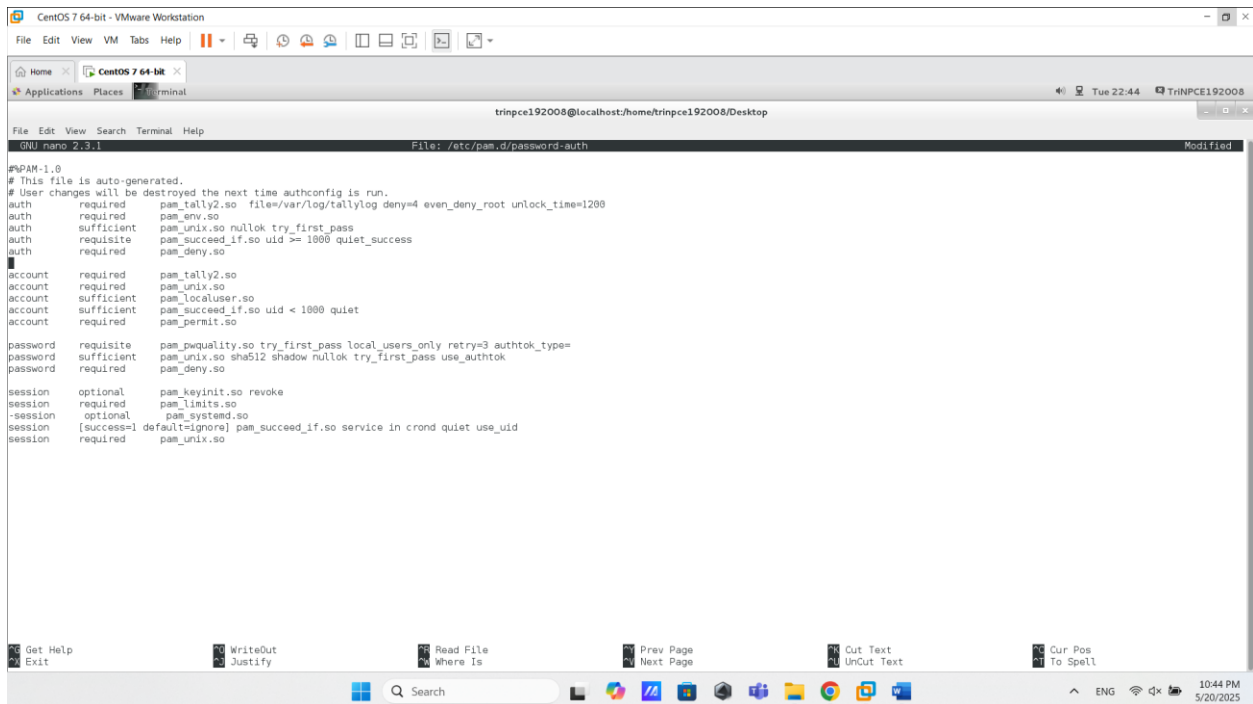
```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 7 64-bit
Applications Places Terminal
trinpce192008@localhost:~/home/trinpce192008/Desktop

[trinpce192008@localhost Desktop]$ su - samson
Password:
Your account has expired; please contact your system administrator
su: User account has expired
[trinpce192008@localhost Desktop]$ clear
[trinpce192008@localhost Desktop]$ su
[trinpce192008@localhost Desktop]$ sudo chage -m 5 -M 90 -i 2 -W 5 samson
[trinpce192008@localhost Desktop]$ sudo chage -l samson
Last password change           : May 20, 2025
Password expires                : Aug 18, 2025
Password inactive               : Aug 20, 2025
Account expires                 : Jul 31, 2025
Minimum number of days between password change : 5
Maximum number of days between password change : 90
Number of days of warning before password expires : 5
[root@localhost Desktop]#
```

ảnh trên: thiết lập thời gian chờ 5 ngày để thay đổi mật khẩu, thời gian hết hạn mật khẩu là 90 ngày, thời gian không hoạt động là 2 ngày và thời gian cảnh báo là 5 ngày

.....

Hands-on lab for configuring pam_tally2:



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 7 64-bit
Applications Places Terminal
trinpce192008@localhost:~/home/trinpce192008/Desktop

File Edit View Search Terminal Help
GNU nano 2.3.1 File: /etc/pam.d/password-auth Modified

#%PAM-1.0
# This file is auto-generated.
# User changes will be destroyed the next time authconfig is run.
auth      required      pam_tally2.so  file=/var/log/tallylog deny=4 even_deny_root unlock_time=1200
auth      required      pam_env.so
auth      sufficient    pam_unix.so nullok try_first_pass
auth      requisite     pam_succeed_if.so uid >= 1000 quiet_success
auth      required      pam_deny.so

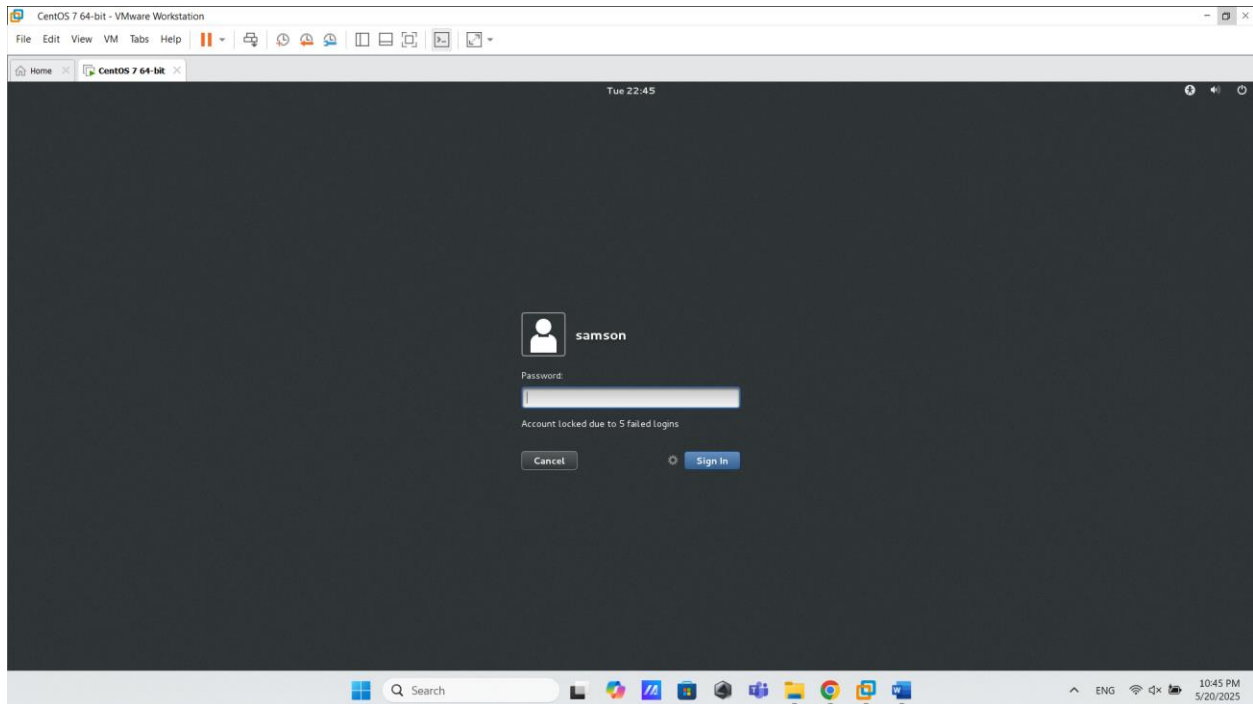
account   required      pam_tally2.so
account   required      pam_unix.so
account   sufficient    pam_localuser.so
account   sufficient    pam_succeed_if.so uid < 1000 quiet
account   required      pam_permit.so

password  requisite     pam_pwquality.so try_first_pass local_users_only retry=3 authtok_type=
password  sufficient    pam_unix.so sha512 shadow nullok try_first_pass use_authtok
password  required      pam_deny.so

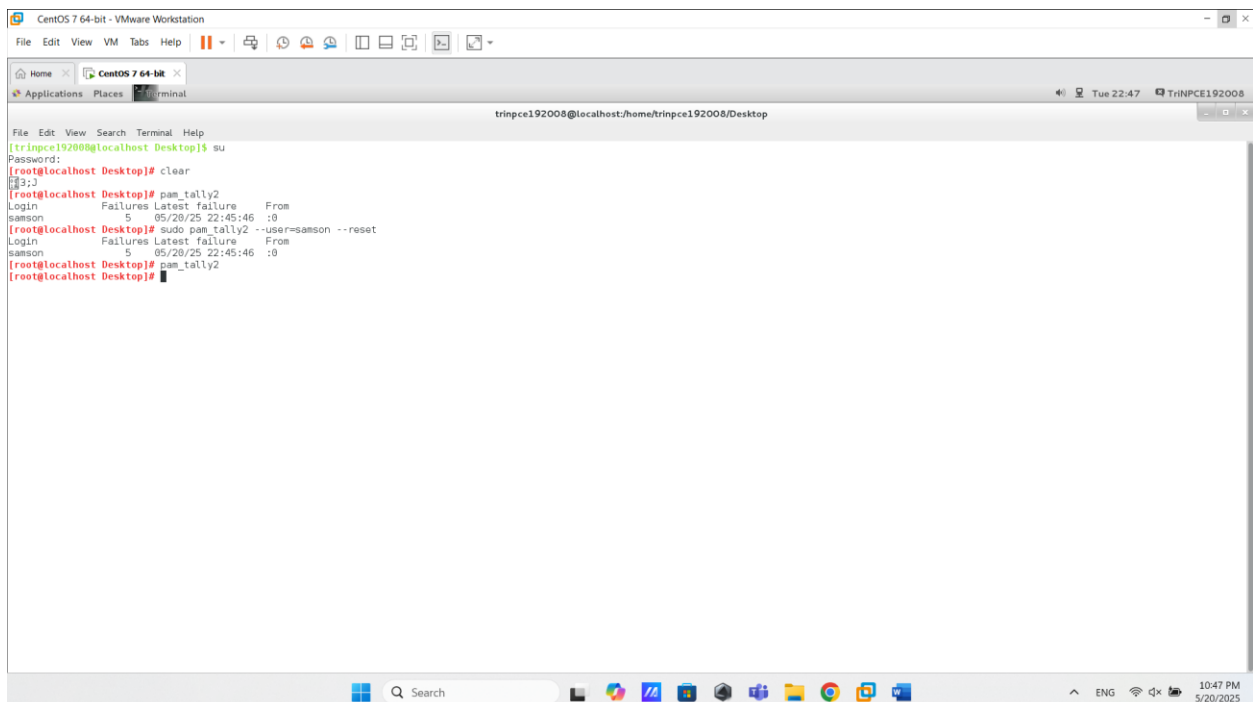
session   optional      pam_keyinit.so revoke
session   required      pam_limits.so
-session  optional      pam_systemd.so
session   [success=] default=ignore pam_succeed_if.so service in crond quiet use_uid
session   required      pam_unix.so

Get Help  WriteOut  Read File  Prev Page  Cut Text  Cur Pos
Exit      Justify  Where Is  Next Page  UnCut Text To Spell
```

ảnh trên: thêm nội dung cho file /etc/pam.d/password-auth để login sai quá số lần sẽ khoá tài khoản user



ảnh trên: nhập thử mật khẩu sai quá 4 lần và tài khoản samson bị khoá



ảnh trên: reset tài khoản cho samson